

Pushkar Jain

+91-9817138143 | pushkrjainn@gmail.com | linkedin.com/in/pushkrjain | github.com/pushkrjain

PROFESSIONAL SUMMARY

AI/ML Engineer building production multi-agent systems on Azure using Azure OpenAI, LangGraph, and FastAPI. Shipped OAuth 2.0 authentication and CI/CD pipelines, cutting latency 35% and cost 40% through async processing. Co-authored an IEEE-published paper on real-time computer vision.

TECHNICAL SKILLS

AI & Frameworks: Azure OpenAI, Multi-Agent Systems, LangChain, LangGraph, RAG, Semantic Search

Cloud & Backend: Python, Azure, Docker, FastAPI, gRPC, REST APIs, Microservices, AsyncIO, Container Apps

Auth & Security: OAuth 2.0, JWT, Google Identity, CORS, Secure Session/Cookie Management

Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems

System Design: Distributed Systems, System Design, Scalability, API Design, Caching, Fault Tolerance, Observability

Data & DevOps: Pandas, NumPy, OpenCV, Vector Search, GitHub Actions (CI/CD), Git, Linux

Agile: Scrum, Sprint Planning, Standups

EXPERIENCE

AI/ML Engineer Intern

Jul 2025 – Present

Docu3c

Newcastle, Washington

- Engineered production multi-agent orchestration systems on Azure using Azure OpenAI, LangGraph, and FastAPI, deploying microservices to Azure Web Apps and Container Apps via GitHub Actions CI/CD.
- Architected a complete OAuth 2.0 authentication system (Google Identity) with a dual-token strategy—60-min access tokens, 3-day refresh tokens via HttpOnly cookies—including refresh, logout, and current-user endpoints.
- Designed and built the FastAPI backend from scratch, including the middleware stack and modular routing across auth, chat, articles, and files.
- Diagnosed and fixed a critical HTTPS downgrade vulnerability via proxy-header trust middleware; implemented async pipelines and fault-tolerant caching, cutting latency 35% and improving cost efficiency 40%.

PROJECTS

Deterministic LLM Video Synthesis Pipeline (Video Forge)

Mar 2026

Stack: *Claude Sonnet 4.5, GSAP, Playwright, FFmpeg*

- Engineered high-throughput distributed pipeline converting natural language to 1080×1920 MP4 videos.
- Achieved 100% first-pass success rate with validation layers (zero retries).
- Implemented parallel processing ensuring byte-identical outputs (SSIM = 1.000).

Multi-Agent AI Orchestration System

May 2025

Stack: *LangGraph, gRPC, FastAPI, Prometheus*

24-hr Challenge

- Reduced average task latency 45% by resolving an inter-agent communication bottleneck via connection pooling and async message queuing.
- Architected a distributed system for agent coordination with fault-tolerant pipelines.
- Developed a CLI tool with real-time observability using Prometheus metrics and custom instrumentation.

AI-Powered Object Detection for Smart Cities

Oct 2024 – Dec 2024

Stack: *YOLOv8n, OpenCV, Pandas*

Published in IEEE CCICT 2026

- Built a real-time AI traffic monitoring prototype using YOLOv8n to detect and classify vehicles, pedestrians, and animals in live video at 30 FPS, with a rolling-window alerting algorithm for automated PDF incident reports.
- Achieved 94.5% average detection precision (97.0% for pedestrians) at a 0.5 confidence threshold; extended the system to multi-threaded dual-camera monitoring.
- Published in IEEE CCICT 2026 (IEEE Xplore), DOI: 10.23919/CCICTCPS00010.2026.00035.

EDUCATION

VIT Bhopal University

Integrated M.Tech in Artificial Intelligence

Sehore, India

Expected May 2028